

Friction: Friend and Enemy

GROW Science · Autumn 1 · Week 3 · Lesson A · 40 minutes

Teach Monday's explanation and shared reasoning only. Stop after We Do 2. Tuesday's 32-minute workshop, 4-minute Lundy response and 4-minute exit are a separate lesson.

Learning and evidence

Objective: explain what friction does, with one helpful and one unhelpful example. Success includes identifying the touching surfaces, predicting a surface comparison and supporting a claim. Monday's adult demonstration is not evidence that the pupil performed Tuesday's test.

Exact source pacing

Elapsed / source stage	Mins	Teaching action
00–01 · Title	1	Name the objective; settle the room.
01–05 · Arrival Task	4	Optional hand rubbing; choose supported, standard or stretch response.
05–07 · Today at a Glance	2	Define friction; judge the job; predict and compare surfaces.
07–16 · I Do 1	9	Model same release twice; change only the surface; invite prediction and observation.
16–26 · We Do 1	10	Start with bike brakes and engine parts. Identify contact, motion and intended job.
26–30 · I Do 2	4	Model the helpful/unhelpful sentence; explain what counts as evidence.
30–40 · We Do 2	10	About 8 min: sort, justify and challenge a changed case. Final 2 min: one Lesson A exit route. Label/save and stop at 40 minutes.

These seven data-timer values total 40 minutes. Tasks, support, optional cards and corrections fit within these stages; they do not create extra lesson time.

Print and prepare

Each pupil: common pages 1–2, one response page (3–5), one arrival page (7–9), organiser page 10 and one exit route (11–13; two copies per sheet). Page 6 has reusable cards. Use screen or paper once; do not require both.

Equipment, access and science

The demonstration is evidence to discuss; models and animations are illustrations.

Demonstration and fair comparison

Use an adult-secured low ramp, a light sliding block, two firmly fixed surface samples (such as fabric and a smooth board), a ruler if comparing stops, and a tray or padded stop. Keep the same block, slope, release mark, no-push release and measuring start. Repeat the same release before changing one planned variable: the surface.

Locate the zero point before recording any distance. Keep the ramp-to-surface join consistent; a raised join or a different push can change the stop. A non-slide is a real observation. Do not invent a distance or guarantee that one surface will always produce the shorter run.

Safety and classroom access

Check the setup against the school's local practical arrangements. Keep the run short, secured and away from edges; keep hands out of the path. No oil, loose powders, flames, heated objects or pupils sliding on floors. Stop if a surface shifts or the run-out is unsafe.

Hand rubbing is optional and brief; stop at discomfort. Offer observing an adult or discussing a known example. Accept pointing, spoken responses, signing, drawing, dictation or adult scribing. Give one instruction at a time, quiet wait time and a pass/private-response option. Do not require physical participation to show understanding.

Core explanation and limits

Friction is a contact force between surfaces that slide, or try to slide, past each other. It opposes that relative sliding. Static friction can stop slipping and can help propel a walker forwards; "always opposite an object's travel" is not a universal rule.

The effect depends on the pair of surfaces and their condition. Roughness alone does not determine every result: smooth rubber and smooth glass can grip. Shorter travel after the same release is evidence of greater resistance in this particular controlled setup, not a direct measurement of frictional force.

Offline fallback

If the apparatus is unavailable, use the editable deck's labelled model and ask pupils to predict and reason. Mark their record "model only"; do not claim a class measurement or practical. No publisher clip is required or included. The optional companion's brake-pad scenario can be described: a pad contacts a turning wheel rim and the wheel slows.

What a good answer includes

Use the answer after an attempt; accept equivalent scientific language.

Arrival and surface prediction

Use the four-question arrival answers on pages 6–8. Accept equivalent responses and uncertainty. Hand rubbing is optional; a remembered example or prediction is not an observation made today.

Prediction: accept a justified prediction, including uncertainty. Look for the actual surface pair, one changed variable and fair controls. Assess observations against what occurred in your demonstration; the original illustration's short/long paths are not measured results.

Helpful / unhelpful examples

Situation	Expected reasoning
Bike brakes	Helpful for slowing: pad–wheel contact resists relative sliding and slows the wheel. In a real stop, tyre–road grip also matters.
Trainers on a wet floor	Shoe–floor friction is helpful for resisting a slip. Water can reduce grip; wet floors are not safe practice surfaces. Do not test this by sliding.
Engine parts rubbing	Unhelpful where unwanted contact causes wear or transfers energy to heating. Judge the particular parts/job, not every friction contact in a machine.
Heavy box across carpet	Unhelpful when the intended job is to slide the box: box–carpet resistance makes it harder. It may be helpful if the job is to keep the box still.
A door handle	Helpful: hand–handle friction provides grip, allowing the handle to be turned. Identify the hand, not just “the handle”.

Model comparison (WAGOLL)

Friction between a bike's brake pad and wheel is helpful because it slows the wheel and transfers energy to heating. Friction between moving engine parts can be unhelpful because it heats and wears the parts.

Highlight contact + effect + intended job. This keeps the source contrast while replacing “converts the movement into heat” with a clearer energy-transfer explanation. Learners may use the simpler “helps it slow down” before adding energy language.

Check the reasoning

The task keeps the original five claims but corrects three over-broad phrasings.

Card / answer	Explanation
A · TRUE	Sliding friction opposes relative sliding at the contact. For a box sliding right on a floor, friction on the box acts left. This does not mean friction is always opposite whole-object travel.
B · FALSE	Smooth surfaces can still have friction. Use the changed case of smooth rubber on smooth glass; ask for evidence about the actual pair.
C · TRUE (qualified)	Oil can reduce sliding friction in suitable lubricated contacts. It does not always do so in every mechanism. This is a reasoning card; do not use oil in class.
D · TRUE (qualified)	Walking in the usual way needs shoe–ground grip. Static friction resists the foot slipping; friction on the foot can act forwards. Avoid a universal statement about all locomotion.
E · NOT A SCIENTIFIC CLAIM	“Bad” is an unexplained value judgement. A useful claim identifies the contact, effect, intended job and evidence; friction can help or hinder.

Supported page

Gap: surfaces. Helpful examples: pad and wheel slow the wheel; hand and handle provide grip. Unhelpful: moving engine contacts heat/wear; box–carpet resistance hinders sliding. Smooth surfaces can grip. Accept pointing and dictated reasoning; record the help provided.

Standard page

On a box sliding right, the friction arrow points left along the contact. The comparison must name the two surfaces and why resistance helps or hinders. A valid counterexample is smooth rubber gripping smooth glass. A polished drawing is not required.

Stretch page

Expect pad–wheel slowing/energy transfer contrasted with machine-contact wear/heating. To test a surface pair fairly, use the same block, load, slope, release, join and measuring start, with repeats. Unknown touching materials need the intended job, motion/attempted sliding and actual surface conditions before “helpful” can be judged.

Useful check: ask a pupil to change one word in a claim and explain how the answer changes. Do not turn this into an extra activity beyond the ten-minute We Do 2 stage.

Close Monday; retain the evidence

Revision 1 · 6 September 2026 · exemplar lesson only

Stop and carry forward

At the end of We Do 2, stop, label and retain the work. Tuesday resumes at Independent Work. Keep Monday's prediction available for Tuesday's separate 40-minute lesson: workshop 32 minutes (4 + 20 + 2 + 6), Lundy response 4 minutes, exit ticket 4 minutes. This exemplar does not contain the complete Tuesday investigation pack.

Record only what happened

Evidence	Adult note / support given
Explained friction at a contact	
Gave a helpful and an unhelpful example	
Predicted / discussed the demonstration	
Own practical performed?	Not inferred from watching a model or demonstration.

Pupil / date: _____

Adult / role: _____

Source and conversion record

Canonical weekly lesson: Science_Teesside/Grow/SCI_G_W3_Friction.html. Current Monday companion: resources/GS_W3A.html. Lessons repository source commit: 5018839965d424afd31bc0ac1b74dc8f713ceca2.

https://github.com/MattRoper1977/Lessons/blob/5018839965d424afd31bc0ac1b74dc8f713ceca2/Science_Teesside/Grow/SCI_G_W3_Friction.html

The seven Monday stage timers and all five helpful/unhelpful situations are retained. Printable tasks map to these stages; the selected response page replaces lengthy copying. Tuesday's results table and exit question are intentionally outside this Lesson A exemplar.

Editorial corrections: "Friction acts against movement" becomes contact-specific sliding friction; "Oil reduces friction" becomes conditional; the walking claim is limited to walking in the usual way; the WAGOLL uses energy transfer and slowing. These corrections match the companion's scientifically qualified model and are not changes to the teaching sequence.

No private OneDrive file is changed; no external clip is copied. The approved silver M has not been substituted or redrawn. These documents use public Made by Matt text branding pending the coordinator's original-logo integration.

Friction arrival

Supported · Four minutes within Lesson A

Allow four minutes within the existing arrival stage. Read or scribe as needed. Start from what pupils show; the word friction is not required before it is taught. Accept uncertainty and equivalent responses. Discuss one useful observation rather than completing a second record. A remembered example or prediction is not a measurement made today.

For question 3, hand rubbing is optional. If comfortable, gently rub your hands for up to five seconds; stop at discomfort. You can watch an adult or describe a familiar hand-rubbing example instead. Say whether you observed it, remembered it or predicted it.

1. A push moves a toy car away. Which action moved it? Push / pull. Push. Accept pointing to the word or showing the action.	2. Look at two palms pressed together. Which surfaces touch? Palm and palm / palm and sleeve. Palm and palm. Accept indicating the two touching surfaces.
3. Use your chosen route. What do you notice when hands rub? Say, point or draw one observation. Accept what was actually noticed, including no noticeable warmth. A remembered or predicted change must be identified as such; do not invent an observation.	4. A shoe needs to grip the ground. Does grip help walking or make slipping easier? Help walking / make slipping easier. Help walking. Grip helps stop the shoe slipping against the ground.

Use the response to choose an explanation or scaffold. No automatic assessment outcome or extra pupil record.

Friction arrival

Standard · Four minutes within Lesson A

Allow four minutes within the existing arrival stage. Read or scribe as needed. Start from what pupils show; the word friction is not required before it is taught. Accept uncertainty and equivalent responses. Discuss one useful observation rather than completing a second record. A remembered example or prediction is not a measurement made today.

For question 3, hand rubbing is optional. If comfortable, gently rub your hands for up to five seconds; stop at discomfort. You can watch an adult or describe a familiar hand-rubbing example instead. Say whether you observed it, remembered it or predicted it.

1. Name the two actions that can change how an object moves: p___ and p___. Push and pull. These are familiar ways of applying forces.	2. When hands rub, name the two touching surfaces. The skin of one palm and the skin of the other palm.
3. Use your chosen route. Describe one change you notice or already know. If you know the force's name, add it. Accept an actual observation or a clearly identified remembered example. Hands may feel warmer. Friction is the optional force name; no noticeable warmth is also a valid observation.	4. A bike brake pad presses on the wheel rim. Predict what happens to the turning wheel. The wheel slows. The pad and rim resist sliding against each other. Treat the response as a prediction unless an actual demonstration is observed.

Use the response to choose an explanation or scaffold. No automatic assessment outcome or extra pupil record.

ARRIVAL / STRETCH / STAFF ANSWERS

Friction arrival

Stretch · Four minutes within Lesson A

Allow four minutes within the existing arrival stage. Read or scribe as needed. Start from what pupils show; the word friction is not required before it is taught. Accept uncertainty and equivalent responses. Discuss one useful observation rather than completing a second record. A remembered example or prediction is not a measurement made today.

For question 3, hand rubbing is optional. If comfortable, gently rub your hands for up to five seconds; stop at discomfort. You can watch an adult or describe a familiar hand-rubbing example instead. Say whether you observed it, remembered it or predicted it.

1. A toy car is pushed, then slows on a level floor. Name one force you think could slow it. Not sure is useful. Accept a plausible resistive force, such as friction in wheel or axle contacts or air resistance. Do not require a single location or teach that friction always opposes the whole object's travel.	2. A hand grips a door handle. Name the touching surfaces and explain what the grip helps you do. Hand and handle. Grip helps turn the handle without the hand slipping over it.
3. Use your chosen route. Suggest why rubbing hands may feel warmer. Give a prediction if you cannot observe it. Work done while rubbing transfers energy to thermal stores, so the skin may warm. Accept an initial idea to revisit; do not claim that every brief rub must produce noticeable warming.	4. Two toy cars start at different speeds on different surfaces. Is that a fair surface comparison? Name one thing to keep the same. No. Starting speeds differ as well as surfaces. Compare using the same car and a consistent release, changing the surface only; accept one sensible control.

Use the response to choose an explanation or scaffold. No automatic assessment outcome or extra pupil record.

Friction exit guidance

Use the last two minutes of W3A's existing ten-minute We Do 2 stage for one exit route. Use about eight minutes for the shared sort, correction and changed case. Stop at 40 minutes. Receive a spoken, signed, pointed, drawn or written response; record only the help actually given. A saved answer or opened explanation does not establish understanding, Audience, Influence or R. Use the response to choose a scaffold or the next explanation. W3B retains its separate 32 + 4 + 4 minute plan.

Supported

1. Left. Friction on this sliding box acts along the contact, against its slide to the right.
2. Yes. Pad–rim friction helps slow the wheel. The touching surfaces and the intended job make this a helpful example.

Standard

1. Surfaces. Accept a clear equivalent that identifies the two contacting surfaces.
2. Slow down or stop. The brake pad and rim resist sliding against one another.

Stretch

1. The pad and rim contact; friction resists their relative sliding and helps slow the wheel.
2. Unwanted rubbing can heat and wear moving parts. Judge friction against the intended job, not as an always-bad force.

The shared knowledge organiser is on pupil page 10. It names contacts, qualifies roughness and energy claims, and separates the model from actual observations.

Next lesson: Friction: test the surfaces